A Mixed Methods Assessment of Family Influence on Weight-Related Behaviors among African-Americans

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Introduction

- African-Americans are disproportionately overweight in the U.S. [1].
- Overweight children are more likely to be overweight and obese as adults, and suffer adverse health outcomes [2, 3].
- Over 55% of African-American women and 37% of African-American men 20 years and older were classified as obese in 2011-12 [4].
- African-Americans tend to lose less weight than whites in weight-loss interventions [5,6].
- Motivation and adherence may play a role [7,8], but interventions targeting the values/customs of African-American culture may be more effective than general interventions [9].
- Factors such as family rules, emotional support, encouragement, positive reinforcement, and family involvement influence the decisions of children and parents in families [10].

Purpose

Conduct an exploratory study using a mixed methods approach to:

- Explore family influences on weight-related behaviors (physical activity and healthy eating)
- Examine relationship between family dynamics and weight-related behaviors

Screening of Participants

Each family will be screened via telephone and must meet the following criteria:

- Self-identify as African-American
- Minimum of two family members
- At least one overweight child between 10-14 years
- At least one obese adult who is viewed as primary caregiver
- All members of the family older than 6 years must participate in focus group

Prior to focus groups, weight status of the overweight child and obese adult will be verified.

Methodology

Recruitment

- Mail recruitment flyers and personalized letters to: churches, community centers, libraries, and community organizations in the Twin Cities
- Appear on *Community Health Dialog with Clarence Jones*, a health-focused show on the local radio station KMOJ 89.9 FM
- Publish print article in the *Minnesota Spokesman Recorder* and *Insight News*
- Use social media, specifically Facebook, Instagram, and Twitter



Figure 1: African-American family

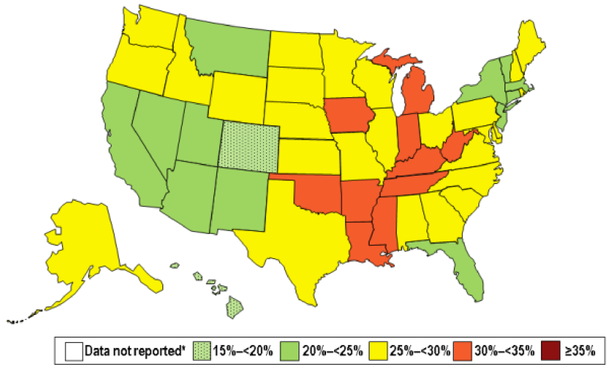


Figure 2: Prevalence of self-reported obesity among white Americans 2011-2013

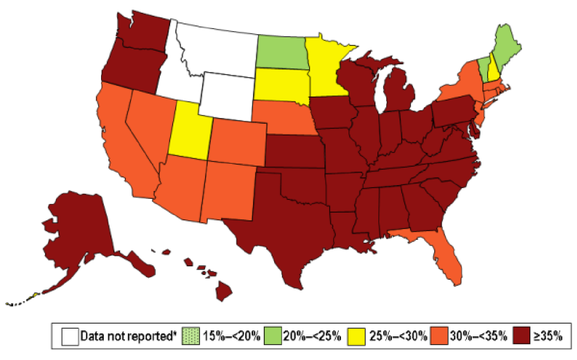


Figure 3: Prevalence of self-reported obesity among black Americans 2011-2013

Qualitative Data Collection

- Focus groups will be conducted in the home of families, lasting 60-90 minutes.
- Refreshments will be offered during focus groups. Each family member will receive $10.
- Focus group participants 10 years and up will complete a short demographic survey.
- Physical Activity and Media Inventory (PAMI) and Home Food Inventory (HFI) will be conducted. Families will receive an additional $10 for completing the PAMI and HFI.
- Upon transcription completion, NVivo will be used to analyze the focus group data.

Quantitative Data Collection

- One overweight child and one obese adult from each family will wear an accelerometer for seven days to collect physical activity data.
- Participants will complete a survey regarding dietary intake, family dynamics, familial social support, and physical activity-related psychosocial measures.
- Participants will receive $10 for completing the quantitative measures.
- Surveys will be entered, cleaned, and analyzed using SAS 9.4.

Timeline

Proposed Study Activities	Timeline: June 2015 – May 2016									
	J	J	A	S	O	N	D	J	F	M
IRB application	■	■								
Development of focus group guide & survey	■	■	■							
Family recruitment			■	■	■					
Pilot test focus group guide & survey				■						
Focus groups, home inventories, & surveys with 10 families					■	■	■			
Focus group transcription & survey data entry/cleaning					■	■	■	■		
Analysis of focus group & survey data								■	■	■

Expected Results

We anticipate the focus groups will yield rich data, concerning the influence of the following factors on eating habits and physical activity behaviors:

- Familial interactions
- Familial social support
- Familial control related to parental rules and monitoring

Results will be used as pilot data for a resubmission of an NIH R01 application that will propose a family-based intervention to increase physical activity in overweight African-American youth.

Selected Citations

Introduction Credits:

1) Wang, Y. and M.A. Beydoun. Epidemiol Rev, 2007. **29**: p. 6-28.
2) Whitaker, R.C., et al. N Engl J Med, 1997. **337**(13): p. 869-873.
3) United States. Dept. of Health and Human Services, 2nd ed. 2000, Washington, DC: U.S. Dept. of Health and Human Services : For sale by the U.S. G.P.O., Supt. of Docs. p. 62
4) Flegal, K.M., et al. JAMA, 2010. **303**(3): p. 235-241.
5) Hollis, J.F., et al. Am J Prev Med, 2008. **35**(2): p. 118-26.

Figure Credits:

1) What is Diabetes? 2015. Retrieved from http://www.veteranshealthlibrary.org/GotoGuides/Diabetes/WhatIsDiabetes.pg
2) Data, Trends, and Maps. 2016. Retrieved from http://www.cdc.gov/obesity/data/prevalence-maps.html